

BENJAMIN HOFFMAN

bhoffman@ethz.ch | [LinkedIn](#) | [Personal Website](#) | Swiss & American Citizen

EDUCATION

ETH ZURICH

Zurich, Switzerland

PhD Student in the Networked Systems Group

Feb. 2025 - present

- ▶ Current research: algorithms, control, and machine learning for large scale systems and networks
- ▶ Supervised by Prof. Laurent Vanbever and affiliated PhD researcher at the ETH AI Center

ETH ZURICH

Zurich, Switzerland

MSc in Robotics, Systems and Control — GPA: 5.7/6.0

Sep. 2022 - Dec. 2024

- ▶ Focus on control and machine learning

LUND UNIVERSITY

Lund, Sweden

Academic Exchange Semester — GPA: 5.7/6.0

Jan. 2022 - Jun. 2022

- ▶ Focus on control and machine learning

ETH ZURICH

Zurich, Switzerland

BSc in Mechanical Engineering — GPA: 5.2/6.0

Sep. 2018 - Aug. 2021

- ▶ Focus on mechatronics and control

KANTONSSCHULE WETTINGEN

Wettingen, Switzerland

Swiss Matura and IB Diploma — GPA: 5.7/6.0 (top 0.5%, salutatorian)

Aug. 2014 - Jun. 2018

- ▶ Major in law and economics, academic exchange semester at the Gymnase de Nyon (VD, Switzerland)

RESEARCH EXPERIENCE

NETWORKED SYSTEMS GROUP — ETH ZURICH

Zurich, Switzerland

Student Research Assistant — part time

May 2024 - Dec. 2024

- ▶ Designed and built a scalable real-world data collection and ABR algorithm evaluation system using Python and Amazon MTurk
- ▶ Performed distributed testing of state-of-the-art streaming algorithms to evaluate their QoE performance in real-world conditions
- ▶ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever

SEMESTER THESIS — NETWORKED SYSTEMS GROUP

Zurich, Switzerland

Machine Learning Based Video Streaming Algorithms — Grade: 6.0/6.0 ([view here](#))

Sep. 2023 - Jan. 2024

- ▶ Researched and evaluated state-of-the-art classical and learning-based ABR video streaming algorithms
- ▶ Developed a network-trace collection and algorithm benchmarking system using Docker, Python, and AWS/Azure SDKs
- ▶ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever

MASTER THESIS — ETH AI CENTER | LEARNING & ADAPTIVE SYSTEMS GROUP

Zurich, Switzerland

Sample Efficient Model-Based Reinforcement Learning — Grade: 5.75/6.0 ([view here](#))

Jun. 2024 - Dec. 2024

- ▶ Using model-based RL for efficient policy learning on a Boston Dynamics Spot quadruped with JAX, PyTorch, and C++
- ▶ Supervised by Jin Cheng, Chenhao Li, Bhavya Sukhija, Lenart Treven, Prof. Stelian Coros, and Prof. Andreas Krause

BACHELOR THESIS — COMPUTATIONAL ROBOTICS LAB

Zurich, Switzerland

State Estimation for Legged Robots — Grade: 6.0/6.0 ([view here](#))

Feb. 2021 - Aug. 2021

- ▶ Researched, developed and implemented different state-of-the-art state estimation approaches using C++
- ▶ Determined the optimal approach still being used in the Lab to date based on an analysis of speed and accuracy
- ▶ Supervised by Dr. Dongho Kang and Prof. Stelian Coros

PUBLICATIONS

- [1] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Strategic Data Collection and Generalization for Learning-Based Adaptive Bitrate Control”. In: *Under Review* (2025).
- [2] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Into the Wild: Real-World Testing for ML-Based ABR”. In: *ACM PACMI* (Seoul, South Korea, 2025). [[PDF](#)]
- [3] **Benjamin Hoffman**, Alexander Dietmüller, and Laurent Vanbever. “Poster: Towards Network Model Generalization using Strategic Data Collection”. In: *ACM SIGCOMM* (Coimbra, Portugal, 2025). [[PDF](#)]
- [4] **Benjamin Hoffman**, Jin Cheng, Chenhao Li, and Stelian Coros. “Learning More With Less: Sample-Efficient Model-Based RL for Loco-Manipulation”. In: *Workshop on Resource-Rational Robot Learning at CoRL* (Seoul, South Korea, 2025). [[PDF](#)]

WORK EXPERIENCE

BAIN & COMPANY Associate Consultant Intern — full time intern (return offer received) ‣ Advised 3 industry leading firms and Fortune 500 companies on performance improvement and IT strategy projects ‣ Conducted empirical research and analyzed large, unstructured datasets to formulate strategic recommendations	Zurich, Switzerland Feb. 2024 - Apr. 2024
ERANEOS Data Science Intern — full time intern (return offer received) ‣ Worked on 5+ projects in finance and the public sector with a focus on ML & GenAI, LLMs, and data analytics ‣ Developed custom LLM solutions and multi-agent systems using fine-tuned local LLMs, RAG, and OpenAI API	Zurich, Switzerland Jun. 2023 - Dec. 2023
GÜDEL GROUP Robotics and Software Engineer — full time intern and part time working student (return offer received) ‣ Machine learning, data science, and software engineering for robotics and industrial analytics ‣ Transformed 3+ quality control procedures to ML-driven processes and rewrote the software for 4+ key applications	Bern, Switzerland Aug. 2021 - May 2023
ETH ZURICH Teaching Assistant for Analysis III and Dimensioning I — part time ‣ Prepared and led weekly exercise sessions for 30+ students, graded assignments, and exam supervision ‣ Elected by students to lead a two-day exam preparation course with 30+ participants for Analysis III	Zurich, Switzerland Sep. 2020 - Jan. 2023

EXTRACURRICULAR EXPERIENCE

UNITED NATIONS – ETH STUDENT TEAM Project Lead ‣ Building a voting-data processing pipeline and a user-facing visualization interface for all 3 main UN bodies ‣ In charge of project management, leading a team of 5 software engineers, and coordinating with UN stakeholders	Zurich, Switzerland Sep. 2025 - present
GLP SIGGENTHAL Board and Committee Member ‣ In charge of communications, as well as planning and implementing campaigns and initiatives on a local level ‣ Member of the advisory body of our municipality on topics of nature and environmental conservation	Obersiggenthal, Switzerland Aug. 2021 - present
VEREIN LEGAL HELP (NGO) Board Member and Head of IT ‣ In charge of strategic initiatives, maintaining long-term growth, as well as all technical and software related projects ‣ Led an organization-wide (100+ users) software transition project resulting in savings of 5000+ CHF annually	Zurich, Switzerland Sep. 2022 - Sep. 2025
ANALYTICS CLUB AT ETH Quants at ETH Member ‣ Growing the quant community at ETH, organizing events of 70+ participants at ETH with leading quant firms	Zurich, Switzerland Feb. 2023 - Dec. 2024
ETH ENTREPRENEUR CLUB Finance Committee Member ‣ Led a project to automate reimbursements, invoicing, and accounting, increasing process efficiency by over 300% ‣ Acted as the principal financial point of contact for leading startups, 50+ members, and major corporate partners	Zurich, Switzerland Sep. 2020 - Feb. 2022

SKILLS & CERTIFICATIONS

PROGRAMMING	Python, C++, MATLAB, JavaScript
FRAMEWORKS	PyTorch, JAX, Pandas, NumPy, SciPy, scikit-learn, Docker, AWS/Azure SDKs, htssim, netUnicorn, Wireshark, tcpdump, OpenAI Gym & API, LangChain, OpenCV, NVIDIA Isaac Sim, Boston Dynamics/Unitree SDK, ROS2
LANGUAGES	English (Native), German (Native), Swiss German (Native), French (B2)
CERTIFICATIONS	J&S Coach Tennis & Snowboard, Certificate in Portfolio and Risk Management (University of Geneva, 2023)

INTERESTS

SPORTS	Competitive tennis player (coach and team captain), snowboarding, running, strong interest in health and fitness
OTHER INTERESTS	Public policy, law and economics, global and local politics, nature and the environment